

The Microbial World

Microbiology

Year 1

College of Medicine

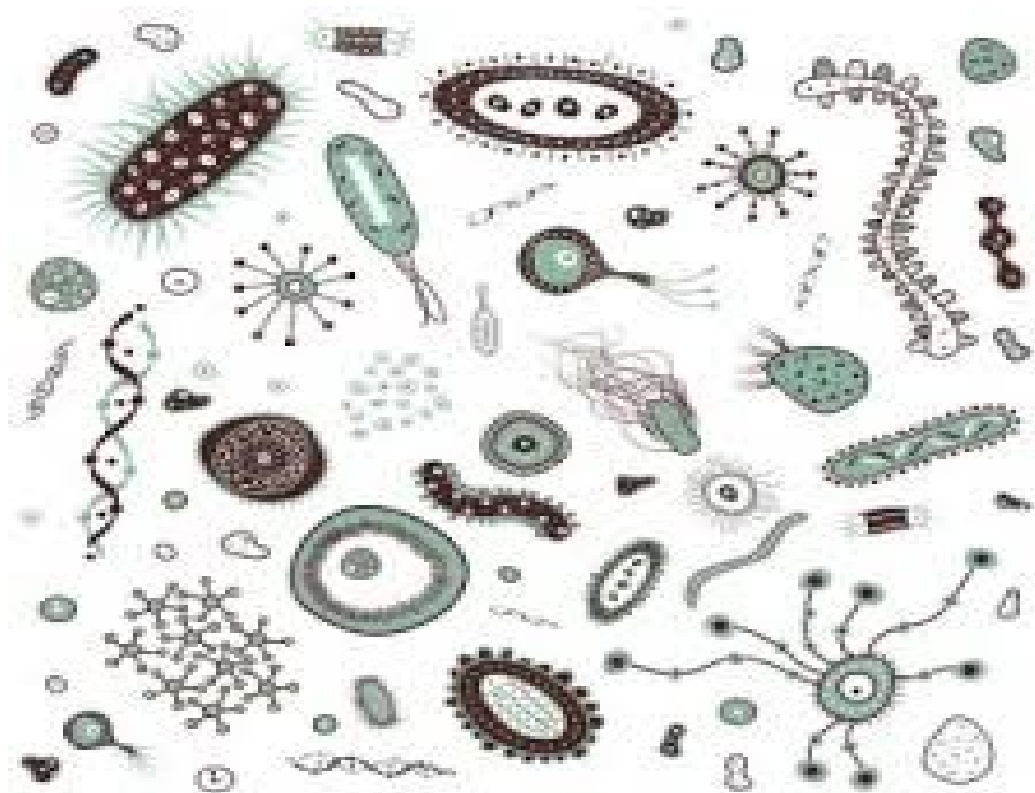
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Microbiology

- Microbiology is the scientific study of microscopic organisms.
- Small living organisms that generally can not be seen with the naked eye



Summary of differences between prokaryote and eukaryote cells

Prokaryotic cells	Eukaryote cells
Small cell ($< 5\mu\text{m}$)	Larger cells ($> 10\mu\text{m}$)
Always unicellular	Often multicellular
No nucleus or any membrane bound organelles	Always have nucleus and membranes bound organelles.
DNA circular, without proteins	DNA is linear and associated with proteins to form chromatin.
Ribosomes are small 70S	Ribosomes are large 80S
No cytoskeleton	Always have cytoskeleton
Motility by rigid rotating flagellum made from flagellin	Motility by flexible waving cilia or flagella made from tubulins.
Cell division is by binary fission	Cell division is by meiosis and mitosis.
Reproduction is always asexual	Reproduction is sexual and asexual.

The importance of microbiology

- **The environment:** they are essential for geochemical cycling and soil fertility.
- **Food:** they are used to produce foods
- **Medicine:** they are used to produce pharmaceutical and industrial compounds.
- **Research:** they are used extensively in research laboratories to investigate cellular processes.
- **Biotechnology:** they have been used to synthesize many medically important substances (chemicals, hormones, proteinsetc)

On the negative side:

They are responsible for the spoilage of food

They cause many diseases of plants, animals, and humans

Medical Microbiology

Medical Microbiology is the study of microorganisms of medical importance:

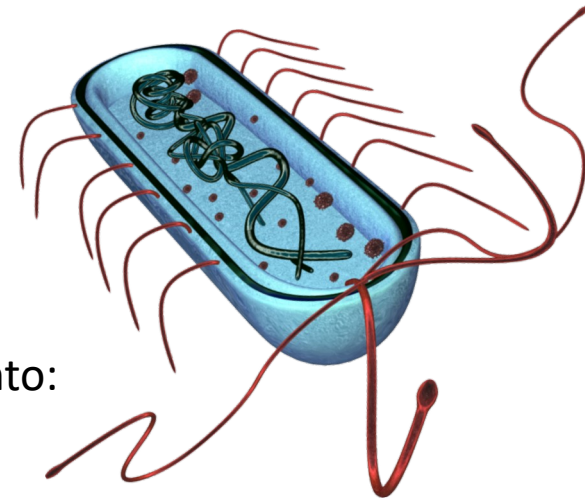
- Bacteria
- Viruses
- Fungi
- Parasites
- Prion



Bacteria

- Microscopic organisms that obtain nutrients from the environments they live on (i.e. human normal flora and pathogens)
- Bacteria increase in number via binary fission
- Majority are useful or harmless. But some cause disease
- **Normal Flora:** Harmless bacteria under normal condition that grow on skin, intestine, oral cavity and other organs
 - They protect our body from invaders and pathogens
 - Helpful in digestion
- **Pathogens:** They cause infection and diseases
- **Opportunistic:** an organism that normally does not harm its host but can cause disease especially when the host's resistance is low

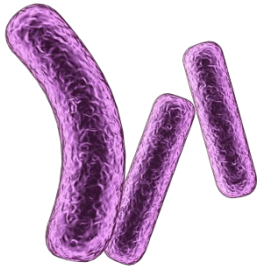
Bacterial structures



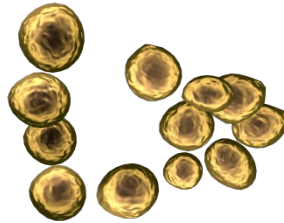
- Essential structures (common to all bacteria)
 - **Cell wall:** (*O antigen*) The basis for bacterial classification into:
 - Gram positive: Stain **blue – violet**
 - Gram negative: Stain **red – pink**
 - **Cell membrane:** Transportation, Secretion, and Energy production
 - **Cytoplasm:** Contains the organelles: ribosome, plasmids...etc
 - **Nuclear material:** Contains the genome of bacteria
- Particular structures (Not in all bacteria)
 - **Capsule:** (*K antigen*) Adhering to host cells and preventing phagocytosis
 - **Spore:** Resistance to heat, dryness, and starvation
 - **Flagella:** (*H antigen*) movement of bacteria
 - **Pili:** Adhering to host cells and transferring genetic materials to other bacteria

Bacterial Shapes

There are 3 main shapes of bacteria



Rods
(Lactobacillus)



Balls or cocci
(Staphylococcus)



Spirals
(Campylobacter)

Growth curve phases

1st: Lag phase

- Bacterial cells adapt to their new environment
- Enzymes and intermediates are formed and accumulate to start the growth

2nd: Exponential phase:

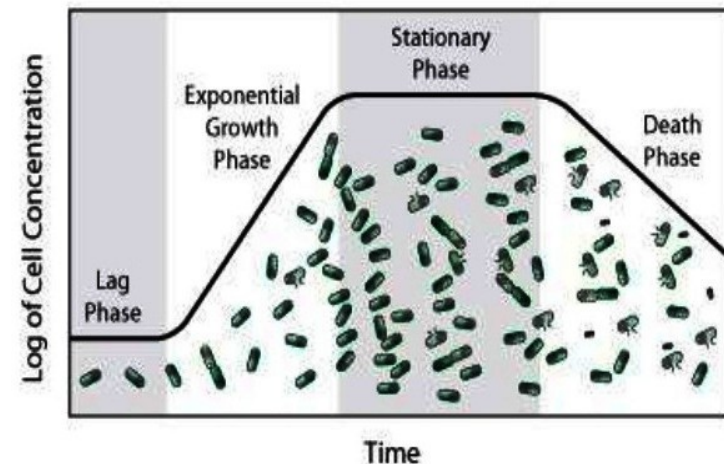
- Constant synthesis of cell materials
- Exponential increase in mass

3rd: Stationary phase:

- A balanced rate of growth to death due to:
 - Exhaustion of nutrients
 - Accumulation of toxic products
- Total cell count increases but viable count stays constant

4th: Decline (Death) phase:

- Increased death rate
- A small number may persist for months or years

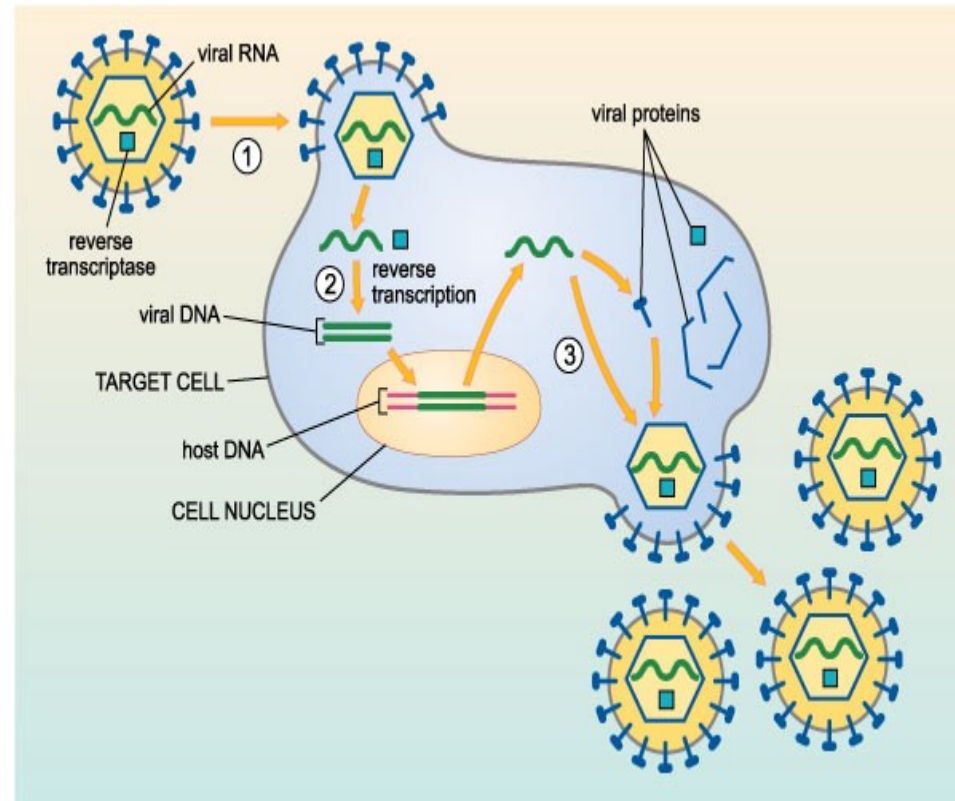


Viruses

Viruses are smaller than bacteria and can sometimes live INSIDE bacteria (Bacteriophage)

Most viruses cause disease (e.g. flu, chickenpox, AIDS)

Viruses infect host cells and use host's reproductive machinery to replicate (i.e. disassemble and reassemble)



Viral Structure

Nucleic acid (Genome)

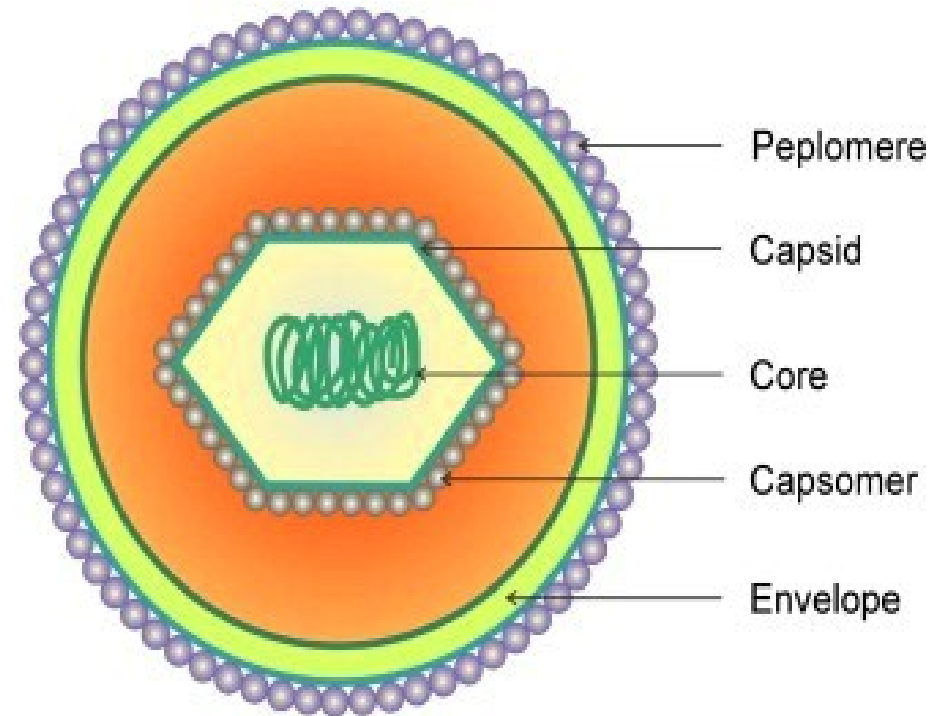
- Either DNA or RNA material

Glycoproteins

- Anchoring the virus to the host cell.
- Transporting genetic material from the virus to the host cell.

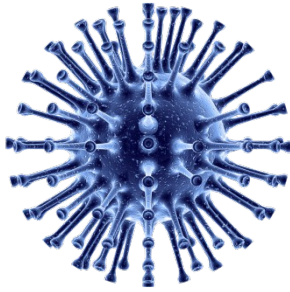
Envelope

- Double lipid layer holding the cells genetic material.
- Not all viruses

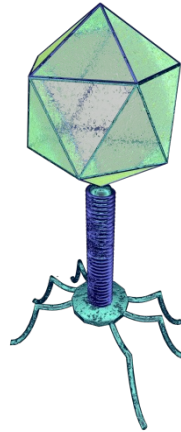


Viral Shapes

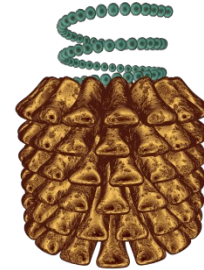
There are 3 different shapes of Virus



Icosahedral
(Influenza)



Complex
(Bacteriophage – a virus which infects bacteria)



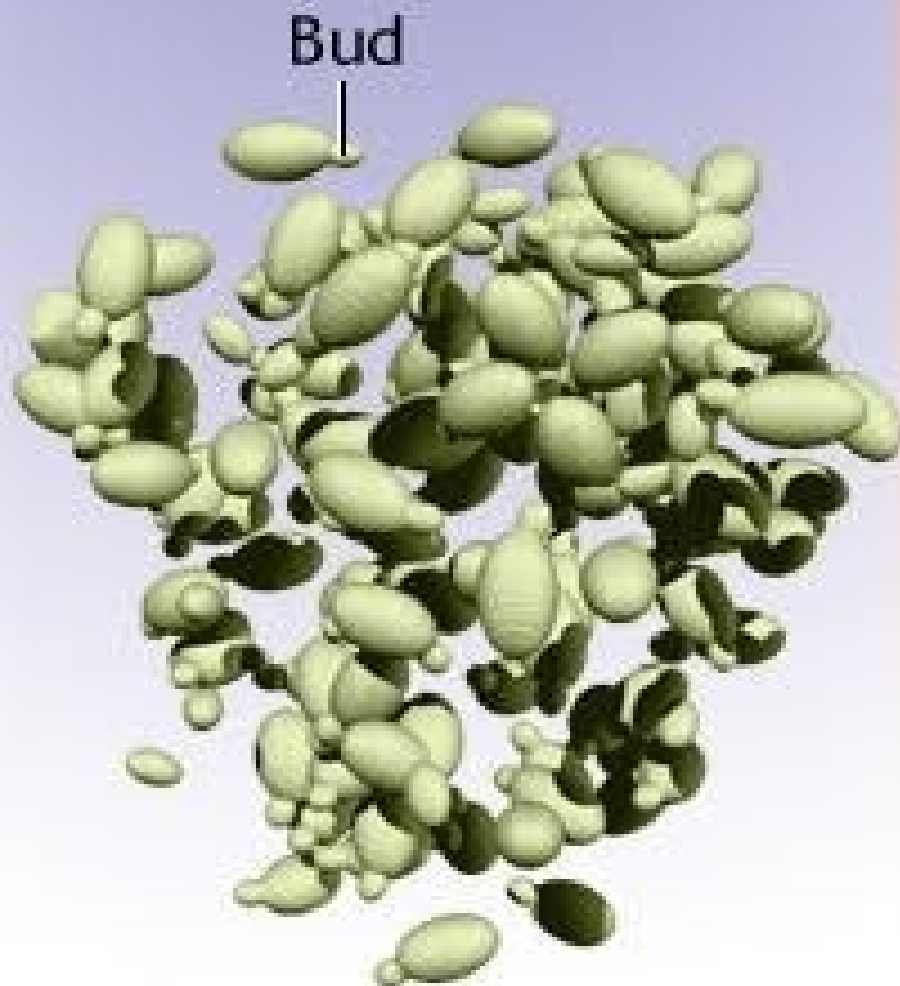
Helical
(Tobacco mosaic virus)

Fungi

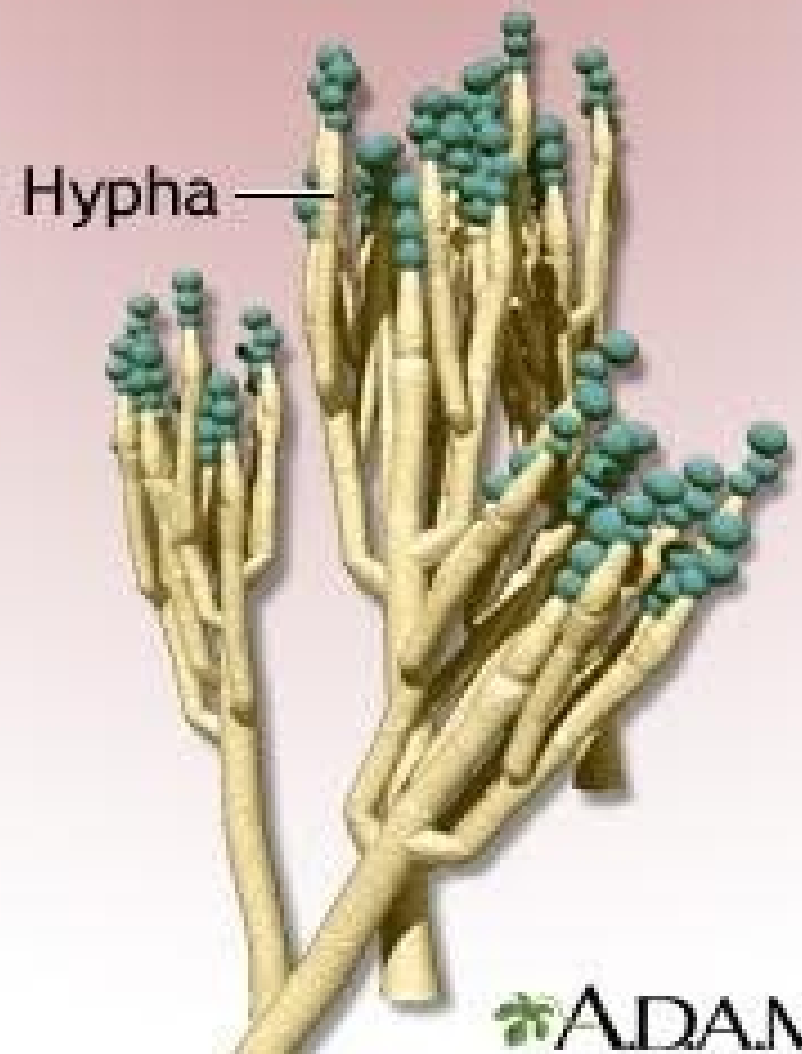
- Eukaryotes similar to plant cells that lack chlorophyll
- The largest and most versatile of all microbes
- Need to absorb nutrients from where they are growing on
- Fungi can be very helpful and humans have used them in
 - The food industry
 - Medicine: making antibiotics
- They can also be harmful to humans:
 - Toxins: mold on bread
 - Destruction of tissues: athlete's foot (caused by dermatophytes)
- Medical important fungi are:
 - **Molds**: A hyphae form (long multicellular structures)
 - **Yeasts**: Unicellular
 - **Mushrooms**: Some are poisonous

FUNGI

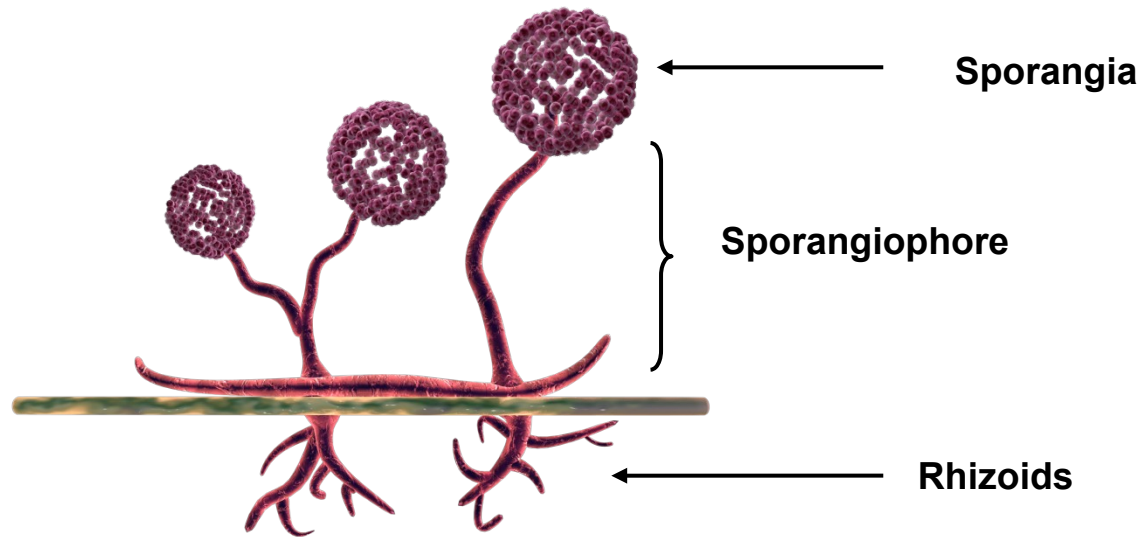
Yeast



Mold



Fungal Structure



Sporangia:

Spore producing body

Sporangiophore:

Filamentous stalk on which the sporangium forms.

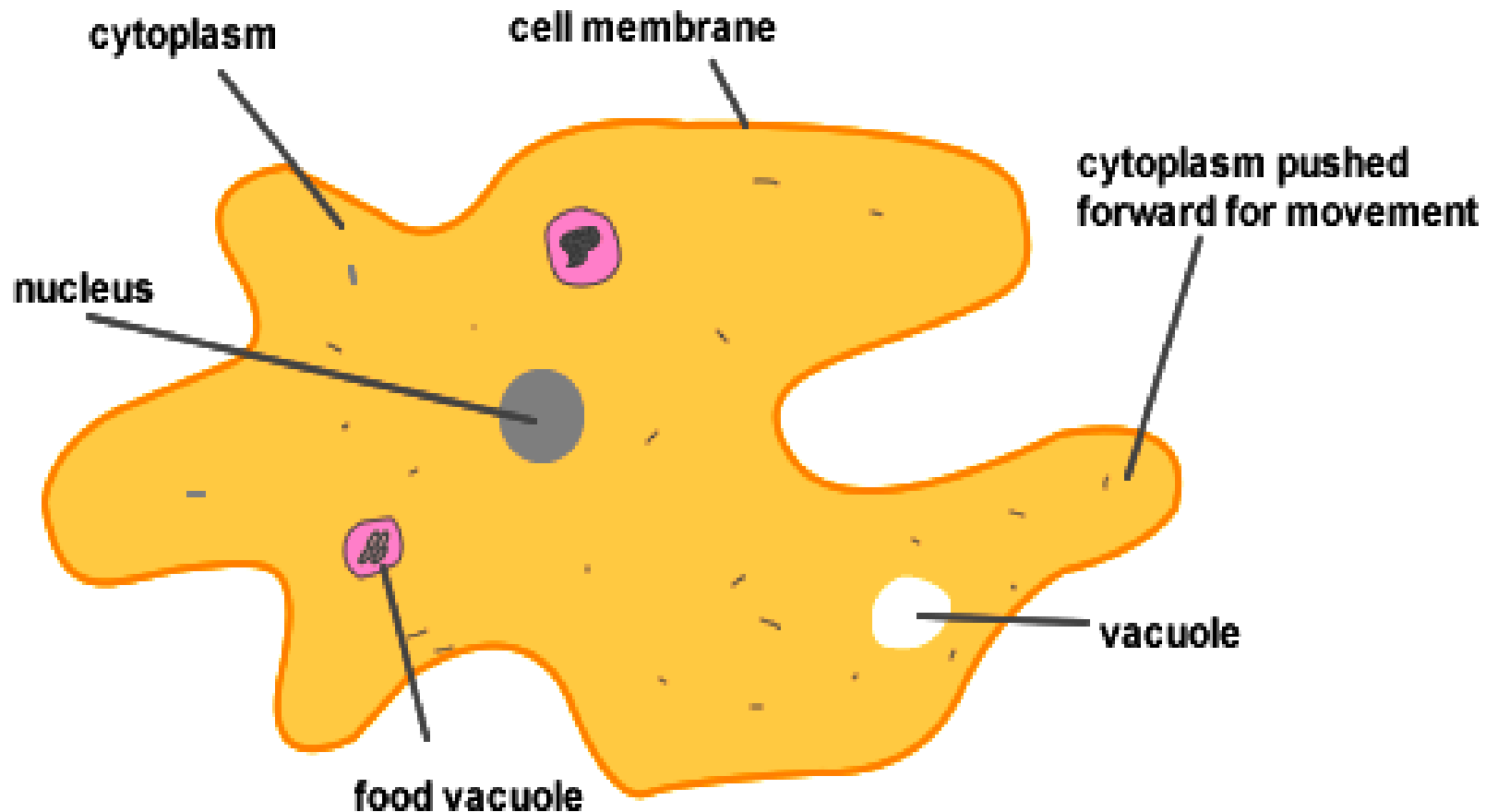
Rhizoids:

The sub-surface **hyphae** are specialized for food absorption.

Parasites

- A parasite is an organism that lives on or in a host and gets its food from or at the expense of its host
- Human parasites are eukaryotic with sexual and asexual reproduction
- Parasites range from microscopic in size to over 30m in length.
 - **Helminths:** worm such as roundworm, pinworm, tapeworm
 - **Ectoparasites:** including mosquito, ticks, lice, and fleas.
 - Attach to the skin and cause disease
 - Vectors or transmitters of other pathogens
 - **Protozoa:** microscopic, one-celled organisms

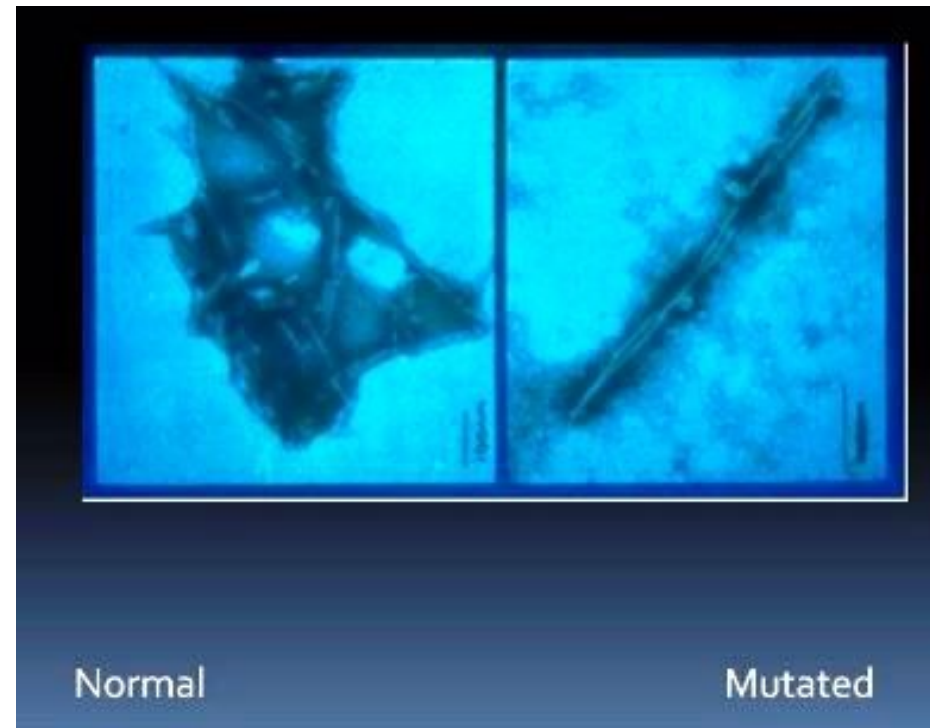
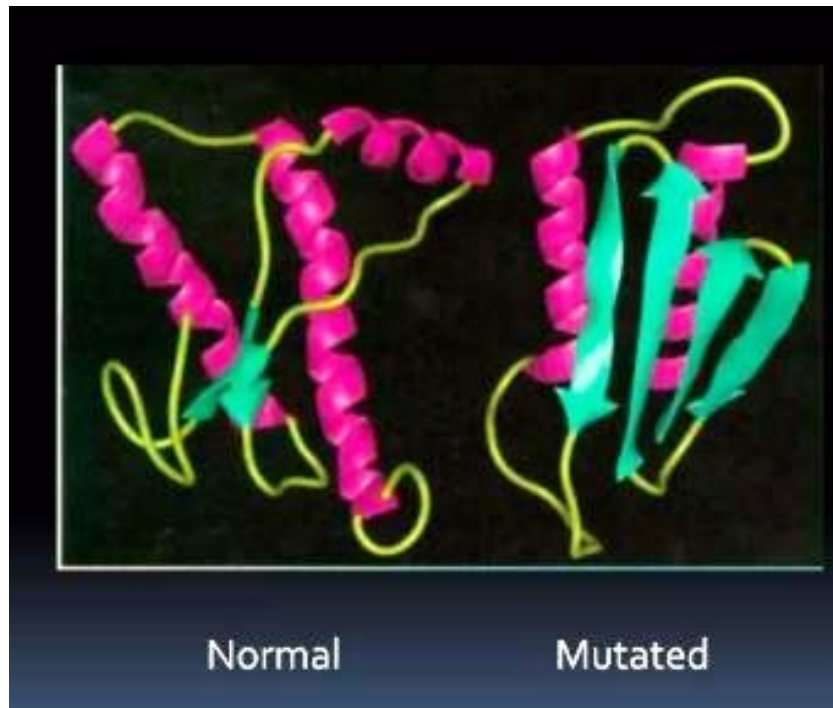
Protozoa (Structure)



Prions

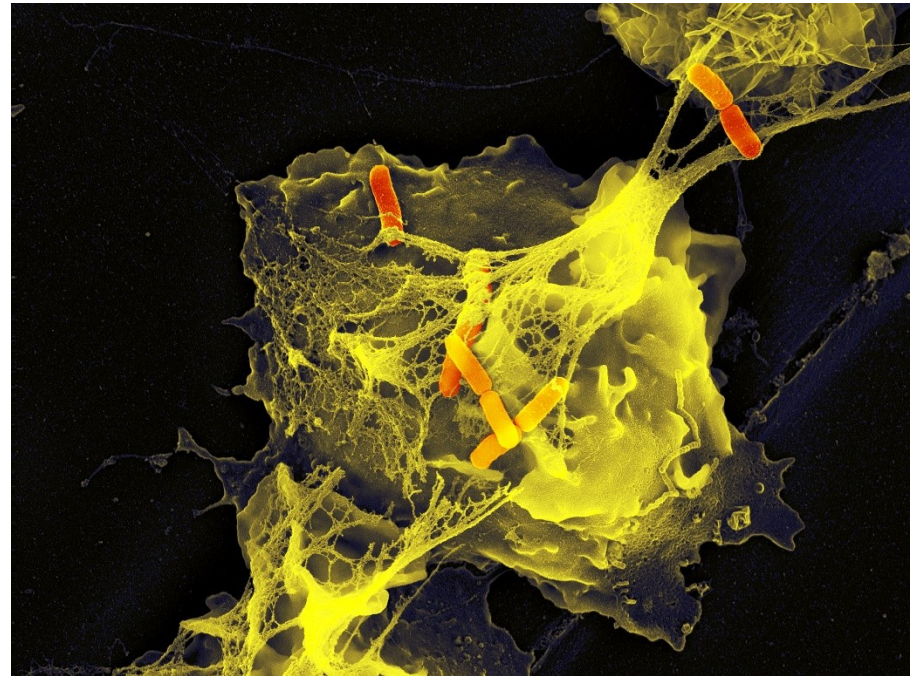
- Abnormal pathogenic protein that are transmissible
- A prion is a misfolded protein that can transmit its property to normal protein and trigger cellular death
- They are able to induce abnormal folding of specific normal cellular proteins (prion proteins)
- Prion proteins are found most abundantly in the brain
 - Creutzfeldt-Jakob disease (CJD)
 - Bovine spongiform encephalopathy (mad cow)
 - Scrapie (neurological disease of sheep)

Prion misfolding



What we should know in Microbiology

1. Names of the diseases (most often with inflammation)
2. Names of the microbes (classification / taxonomy)
3. Type of microbe:
 - Pathogenic microbes
 - Commensal microbes
 - Opportunistic microbes
4. Mode of transmission
5. Treatment and prevention



Mycobacterium tuberculosis (red bacilli) trapped within neutrophils extracellular nets in the lungs of TB patient